

ON SENSE AND REFERENCE

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PHI205

This is a reading guide. Read this first. It tells you what the assigned reading is, where to find it, and offers some guidance on how to approach it and what I am hoping you will get out of it.

This is our first “real” reading assignment, so I will be giving a lot of guidance.

GETTING THE PAPER

The assigned reading is Frege (1948) . This is a translation of Frege (1892). If you read German, you can read the original rather than the translation.

The link I have given you to Frege (1948) is a “DOI” link. “DOI” stands for “Digital Object Identifier.” When you click on the link, it will take you to JSTOR, which is a large repository of journal articles.

It isn’t free to read. You need a subscription. Happily, Milner Library has a subscription.

SAVING THE PAPER

You can read the article online, but I highly recommend that you download the PDF, save it in a folder on your device, and read it from there.

PRINTING THE PAPER

Actually, what I really recommend is that you print it out. It is easier to mark up a print out, and writing your notes by hand in the margins gives you several degrees of freedom that go missing when you try to take notes on a device. That freedom means that it is easier to “think on paper.”

A printout is also easier to use in-class, as we discuss the paper. It is easier to flip back and forth between different pages. It is easier to scribble additional notes and thoughts on the printed-out paper.

And, on writing days, you are not allowed to use your electronic devices. But you are allowed to bring and use notes and printouts. A shambolic printout, covered in a record of your thinking process, will serve you well.

READING THE PAPER

Don’t read the entire article.

When you reach the paragraph, on p. 218 (p. 36 in the original German), that begins “We are thus led to consider subordinate sentences or clauses,” skip to the end (p. 230; p. 50 in German), and read the last two paragraphs, beginning with “Let us return to our starting point!” We are skipping the 12 page discussion of subordinate clauses.

HOW TO READ PHILOSOPHY

Many of you have already taken philosophy classes. So this is not your first time reading a work in philosophy. For some of you it is.

This is a famous paper. Generations of philosophy students have cut their teeth on it.

It is also a hard paper. It is dense. The arguments are not always easy to follow. The puzzles and problems might not be, at first blush, the puzzles and problems you would focus on.

It is your job to roll with it. Read slowly and carefully. Try to figure out what is going on. Pay attention when he gives examples. Try to figure out what the examples are meant to show. Pay attention when he makes distinctions, and how explains them. Pay attention when you think you understand—jot down a note explaining it! And pay attention when you are confused—jot down a note explaining what you are confused about. These notes are invaluable.

Give yourself grace. You will be confused. It will be frustrating. It is okay to take breaks. But this is what you signed up for. And it is not impossible. As mentioned above, generations of philosophy students have done this before you.

One common piece of advice: begin by reading the first page, and then read the last page. That might give you some sense of what the paper is about, and, hopefully, some sense of what the author thinks they succeed at showing. Armed with this, read the rest of the paper.

Another common piece of advice: begin by reading the entire paper through, quickly. Take a few notes, especially notes about the parts you don't understand, but don't get bogged down. That might give you a rough and general sense of what the paper is about (of course, your rough sense might be completely wrong, since you didn't read very carefully). Armed with this, re-read the paper from the beginning, slowly, and figure out what is actually going on.

A third piece of advice: argue with the paper as you read it. Take notes that reflect this. Note where you agree or disagree. Note where you think something smells fishy. Come up with your own examples and try out counterexamples. Note objections as you think of them. Note when an objection you thought of earlier is addressed later. Note when something the author says shifts your way of thinking. Reading slowly is about slowing your mind down. It is about giving your mind time to push and press and probe and reconstruct and consider and reconsider and....

A universal piece of advice: don't expect to get everything you need to get out of this paper on the first reading. Your first reading puts you in a position to try to understand it. Lecture and discussion will put you in a position to see better what you did and didn't understand. Armed with that, you can re-read the entire paper, or re-read selected passages, and new insight will open up. Understanding is a process. Expect it to take time.

Enough advice. Let's get reading.

READING NOTES

I am going to walk you through the first page, sentence by sentence. I won't be doing this for every reading. But we are just getting started, and this reading is dense, and I want to give you a model of what I mean by "read slowly and carefully."

Identity¹ gives rise to challenging questions which are not easy to answer.

¹ I use this word strictly and understand “ $a = b$ ” to have the sense of “ a is the same as b ” or “ a and b coincide.”

What do you think Frege means by ‘identity’? The fact that he provides a footnote suggests that he was worried that you might misunderstand.

Is it a relation?

Here are some examples of relations:

- The *greater than* relation: ‘ $4 > 3$ ’, $4 > 5$.
- The *worse than* relation: ‘This pasta is *worse than* the pasta you made last night.’
- The *is a part of* relation: ‘My nose *is a part of* my face.’

What are some other examples of relations? Here are some identity claims:

- $2 + 2 = 4$
- Dwayne Johnson *is identical to* the Rock.
- Daniel Radcliffe *is identical to* Elijah Wood.

Can you think of reasons someone might have for thinking that identity *is not* a relation?

A relation between objects, names or signs of objects?

Huh. What are these two options? Why are they the two options we should consider?

In my *Begriffsschrift* I assumed the latter.

The *Begriffsschrift* (“Concept Notation”) (Frege 1879; English translation Frege 1967) is a booklet on logic. In it, Frege invents the logic that we now teach in intro logic courses, displacing traditional Aristotelian term logic.

In §8 of that booklet, “Identity of content”, he says,

Whereas in other contexts signs are merely representatives of their content [...] they suddenly display their own selves when they are combined by means of the sign for identity of content; for it expresses the circumstance that two names have the same content. (Frege 1967, 20)

Returning to our reading, he continues,

The reasons which seem to favor this are the following:

Ah, good. We won’t need to read the *Begriffsschrift*, because he is about to explain his reasons to us here. (Notice the use of ‘seem’. That suggests he has changed his mind.)

$a = a$ and $a = b$ are obviously statements of differing cognitive value; ...

Note that Frege does not follow our contemporary convention for use-mention. Following this convention, we would surround the two identity statements, which are here being mentioned, not used, in single quotes:

' $a = a$ ' and ' $a = b$ ' are obviously statements of differing cognitive value; ...

I will not be "fixing" Frege's use-mention to fit this convention. But you should pay attention to it for yourself.

He does not explain what he means by "cognitive value". "Cognitive" means, roughly, "related to thought". So "thought value"? Maybe this will be clearer in light of the reason he offers in support:

$a = a$ holds *a priori* and, according to Kant, is to be labeled analytic, while statements of the form $a = b$ often contain very valuable extensions of our knowledge and cannot always be established *a priori*.

Here Frege uses some philosophical terminology and name-drops Kant. He assumes anyone reading this article has already read Kant ([1781] 1998). He's probably wrong about that in your case. You should look up what '*a priori*' and 'analytic' mean (hint: here 'analytic' is being used in contrast to 'synthetic'). And you should put Kant ([1781] 1998) on your summer reading list!

Did you look up what '*a priori*' and 'analytic' mean? Do so before reading further.

Now you know that Frege is claiming that statements of the form ' $a = a$ ' can be known independently of experience (that is, "hold *a priori*"), and are, at least according to Kant, true in virtue of the meanings of the words and how they have been put together (that is, "analytic"), while statements of the form ' $a = b$ ' are often both synthetic and *a posteriori*—in order to know that $a = b$, you need to have empirical evidence, not just know what the words mean and how they have been put together.

Does all of that seem right to you? Try to think about specific examples. Do you agree that,

- Dwayne Johnson = Dwayne Johnson,

is both *a priori* and analytic? Do you agree that,

- Dwayne Johnson = the Rock.

is neither analytic nor *a priori*?

The discovery that the rising sun is not new every morning, but always the same, was of very great consequence for astronomy. Even today, the identification of a small planet or a comet is not always a matter of course.

That is, the statement,

- The sun that rises today = the sun that rose yesterday,

is neither analytic nor *a priori*, but instead expresses an empirical scientific discovery.

Let's stop to recap. Reading from the top:

Identity gives rise to challenging questions which are not easy to answer. Is it a relation? A relation between objects, names or signs of objects? In my *Begriffsschrift* I assumed the latter. The reasons which seem to favor this are the following: $a = a$ and $a = b$ are obviously statements of differing cognitive value; ... $a = a$ holds *a priori* and, according to Kant, is to be labeled analytic, while statements of the form $a = b$ often contain very valuable extensions of our knowledge and cannot always be established *a priori*. The discovery that the rising sun is not new every morning, but always the same, was of very great consequence for astronomy. Even today, the identification of a small planet or a comet is not always a matter of course.

See where we are? Frege has asserted that identity is either a relation between objects or between names of objects, and he is in the middle of giving us a reason for thinking it is not a relation between objects. So far, he has argued that ' $a = a$ ' and ' $a = b$ ' differ in cognitive value. But it remains to be seen why that should entail that identity is not a relation between objects.

Now if we were to regard identity as a relation between that which the names " a " and " b " designate, it would seem that $a = b$ could not differ from $a = a$ (i.e., provided $a = b$ is true). A relation would thereby be expressed of a thing to itself, and indeed one in which each thing stands to itself but to no other thing.

Frege's argument here is that identity cannot be a relation between objects, because, if it were, ' $a = a$ ' and ' $a = b$ ' would have the same cognitive value. Why? Because both ' $a = a$ ' and ' $a = b$ ' would express the same relation—a relation that holds between a thing and itself, and not between that thing and anything else.

(Notice that this helps clarify the kind of 'identity' Frege is interested in here. *Qualitative identity*—the kind of identity-as-similarity that holds between "identical" twins—is a relation that can hold between one thing and another thing. *Numerical identity*—being one and the same thing—is the relation that holds between each thing and itself, but not between that thing and anything else.)

We have been given an argument. It is good practice, when reading philosophy, to try to reconstruct the arguments you are given.

- NO1. ' $a = a$ ' and ' $a = b$ ' differ in cognitive value.
- NO2. If identity were a relation between objects, ' $a = a$ ' and ' $a = b$ ' would express the same relation.
- NO3. If ' $a = a$ ' and ' $a = b$ ' expressed the same relation, they would have the same cognitive value.
- NO4. So, identity is not a relation between objects.

It is also good practice to immediately start trying to think of objections. This isn't because we are playing some obnoxious game of "Gotcha!" It's because, by trying think of objections, you force yourself to actually pay attention to what is being said. Students often read philosophy in "vibe" mode, and come away with a feeling of understanding, but no actual understanding. You have to fight this tendency.

So, what do you think? Is the argument just reconstructed valid? That is, if you were to grant the premises, would the conclusion follow? Assuming it is valid, are there any premises that look fishy to you? If not, imagine you are a really disagreeable reader, searching for things to disagree with. Where might you poke or prod?

If you just tried to do all of that, you may have run into the problem that the argument is pretty abstract, so it is hard to get real purchase on it. When that happens to you, always try to reach for examples. I've given you the 'Dwayne Johnson' and 'The Rock' examples. Try plugging those into the argument, in place of the abstract ' $a = a$ ' and ' $a = b$ '.

Also, it is fine if you don't come up with any great objections. The point is to try. The goal isn't a "Gotcha!" The goal is to actually understand the argument.

What is intended to be said by $a = b$ seems to be that the signs or names " a " and " b " designate the same thing, so that those signs themselves would be under discussion; a relation between them would be asserted.

Ah, now we are moving on to the second option: if identity is not a relation between objects, it must be a relation between names. What relation, exactly? Co-reference: when you learn that Dwayne Johnson is the Rock, what you have learned is that the name, 'Dwayne Johnson', and the name, 'The Rock', both refer to the same thing. This is a relation between the two names: they co-refer.

But this relation would hold between the names or signs only insofar as they named or designated something. It would be mediated by the connection of each of the two signs with the same designated thing. But this is arbitrary. Nobody can be forbidden to use any arbitrarily producible event or object as a sign for something. In that case the sentence $a = b$ would no longer refer to the subject matter, but only to its mode of designation; we would express no proper knowledge by its means. But in many cases this is just what we want to do.

Oh, shit. Now we seem to have an argument against the view that identity is a relation between names. Let's try to reconstruct it.

- NN1. The connection between a name and its referent is arbitrary and conventional.
- NN2. So, the fact that ' a ' and ' b ' co-refer is arbitrary and conventional.
- NN3. But sentences of the form ' $a = b$ ' can express proper knowledge.
- NN4. Knowledge of arbitrary and conventional facts is never proper knowledge.
- NN5. So, sentences of the form ' $a = b$ ' do not express co-reference.

I've done a fair bit of work for you here, unpacking the argument from the text. I may have gotten it wrong. So, once again, you should be thinking of possible objections, and thinking about which premise(s) seem most dubious. But also, you should be comparing what I've given you with the original text, and thinking about whether my reconstruction is misleading.

If the sign " a " is distinguished from the sign " b " only as object (here, by means of its shape), not as sign (i.e., not by the manner in which it designates something), the cognitive value of $a = a$ becomes essentially equal to that of $a = b$, provided $a = b$ is true.

This sentence is hard to parse. It helps a little if we move the last bit, 'provided $a = b$ is true', to the beginning:

- If ' $a = b$ ' is true, and if each name is distinguished from the other only by its shape, not by the "manner in which it designates something", then ' $a = b$ ' has the essentially the same cognitive value as ' $a = a$ '.

Frege's first argument suggested that, when $a = b$ is true, the relation between the object a and the object b is the same as the relation between the object a and the object a , so nothing at the level of objects can explain a difference in cognitive value.

His second argument suggests that the fact that two signs of two different shapes have both been connected to the same object is arbitrary and conventional. That gets us a difference, but not the right sort of difference. The difference we are after needs to be a difference that can amount to proper knowledge: again, think of

the case of the astronomer who discovers that comet observed 100 years ago is identical to the comet she observes today.

A difference can arise only if the difference between the signs corresponds to a difference in the mode of presentation of that which is designated.

Here, Frege introduces a third potential source of a difference in cognitive value: not the difference in names, and not the difference in objects, but the difference in “modes of presentation” by which each name designates the object.

But what is a “mode of presentation”? What follows is an example, illustrating the idea:

Let a , b , c be the lines connecting the vertices of a triangle with the midpoints of the opposite sides. The point of intersection of a and b is then the same as the point of intersection of b and c . So we have different designations for the same point, and these names (“Point of intersection of a and b ,” “Point of intersection of b and c ”) likewise indicate the mode of presentation; and hence the statement contains true knowledge.

Construct a diagram to help yourself make sense of this example.

It is natural, now, to think of there being connected with a sign (name, combination of words, letter), besides that to which the sign refers, which may be called the referent of the sign, also what I would like to call the *sense* of the sign, wherein the mode of presentation is contained. In our example, accordingly, the referents of the expressions “the point of intersection of a and b ” and “the point of intersection of b and c ” would be the same, but not their senses. The referent of “evening star” would be the same as that of “morning star,” but not the sense.

This is Frege’s famous view about how names work: each name *expresses* a sense, which *determines* a referent. The difference in cognitive value between ‘Dwayne Johnson is Dwayne Johnson’ and ‘Dwayne Johnson is the Rock’ is due to the fact that ‘Dwayne Johnson’ and ‘The Rock’ each express a different sense—a different “mode of presentation”—even though those two different senses determine the same object.

Here ends this reading guide, but not your reading assignment. You still have several pages to go. In those pages, Frege will clarify his view that He then extends that view from names to entire sentences: each sentence *expresses* a sense, which *determines* a referent. In the case of a sentence, the sense is a “thought”, and the referent is a truth value. And he discusses what happens when a name has a sense but lacks a referent, and how that affects the sense and reference of a sentence containing that name.

Try to use this guide as a model. Read slowly. Take each sentence as it comes, and try to figure out what it is saying. Obviously, in writing this guide, I was cheating. I’ve read this paper many times, so I already know what is going on. You are reading this paper for the first time. You will have to read ahead and circle back, and you won’t always be able to make sense of what is going on. That’s all okay! Stick with it and push through. You cannot understand a paper because you’ve read it several times without reading it for the first time, without completely understanding it.

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